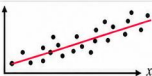
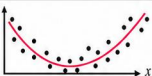


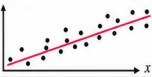
QUICK REFERENCE TO REGRESSION METHODS IN DATA SCIENCE

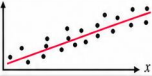
Models, intuition, and shape of the fit

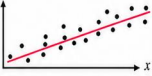
- 1 Linear regression**
 Models a linear relationship between x and y .

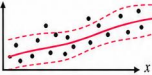
$$y = \beta_0 + \beta_1 x + \varepsilon$$

- 2 Polynomial regression**
 Extends linear regression with powers of x .

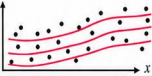
$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \dots + \beta_d x^d + \varepsilon$$

- 3 Ridge regression**
 Penalizes large coefficients with L2 regularization.

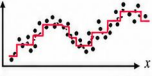
$$\min_{\beta} \|y - X\beta\|_2^2 + \lambda \|\beta\|_2^2$$

- 4 Lasso regression**
 Uses L1 regularization and can select variables.

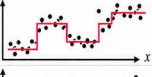
$$\min_{\beta} \|y - X\beta\|_2^2 + \lambda \|\beta\|_1$$

- 5 Elastic Net regression**
 Combines L1 and L2 regularization to stabilize and select variables.

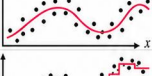
$$\min_{\beta} \|y - X\beta\|_2^2 + \lambda \left[\alpha \|\beta\|_1 + (1 - \alpha) \|\beta\|_2^2 \right]$$

- 6 Bayesian regression**
 Incorporates prior information and estimates a posterior distribution of the parameters.

$$p(\beta | y, X) \propto p(y | X, \beta) p(\beta)$$

- 7 Quantile regression**
 Estimates conditional quantiles (median and percentiles) instead of the mean.

$$\hat{Q}_{\tau}(\tau | x) = x^T \beta_{\tau}$$

- 8 k-nearest neighbors regression (k-NN)**
 Predicts using the average of the k nearest neighbors.

$$\hat{y}(x) = \frac{1}{k} \sum_{i \in N_k(x)} y_i$$

- 9 Decision tree regression**
 Divides the space into regions and predicts piecewise.

$$\hat{y}(x) = \sum_{m=1}^M c_m 1(x \in R_m)$$

- 10 Random forest regression**
 Averages multiple trees to improve robustness.

$$\hat{y}(x) = \frac{1}{T} \sum_{t=1}^T \hat{y}_t(x)$$

- 11 XGBoost regression**
 Combines sequential trees, correcting previous errors.

$$\hat{y}(x) = \hat{y}_0 + \sum_{m=1}^M \eta f_m(x)$$

- 12 Neural network regression**
 Learns complex, non-linear relationships.

$$\hat{y} = f_{\theta}(x)$$


— Pink: Prediction/Fit

● Black: Observed

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